
Presentation 2

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Connection from Last Presentation

Previous Categorization

Time-related Redundancies

- Long time at a single prompt
- Long overall call time

Iteration-related Redundancies

- Repeating same node
- Move to previous node
→ return to same node
- Move to previous node
→ return to different node
- Cycles



New Categorization

Categorize different types of people

1. Listens to all the options and goes to the right menu
2. Listens to all the options, and goes to the wrong menu and goes back
→ **Issue with verbiage**
3. Makes a choice before listening to all the options, and goes to the right menu
4. Makes a choice before listening to all the options, and goes to the wrong menu and goes back → **Issue with the order in which the option is being presented and / or verbiage**

Goal of this presentation



Define Key Assumptions



Identify Repeated Processes in Main Menu & Legal Menu



Identify Prompt Repetition

Key Definitions / Algorithm

"Goes to the right menu"

Definition

- The caller does not go back to the parent node → indication of reaching the intended node

Algorithm

- Multiple non-consecutive instances of the same "Activity Name"

"Fully Listened To"

Definition

- The caller remained in the prompt until the second last option

Algorithm

- Compare prompt length vs duration of call at that prompt node

"Successful Call"

Definition

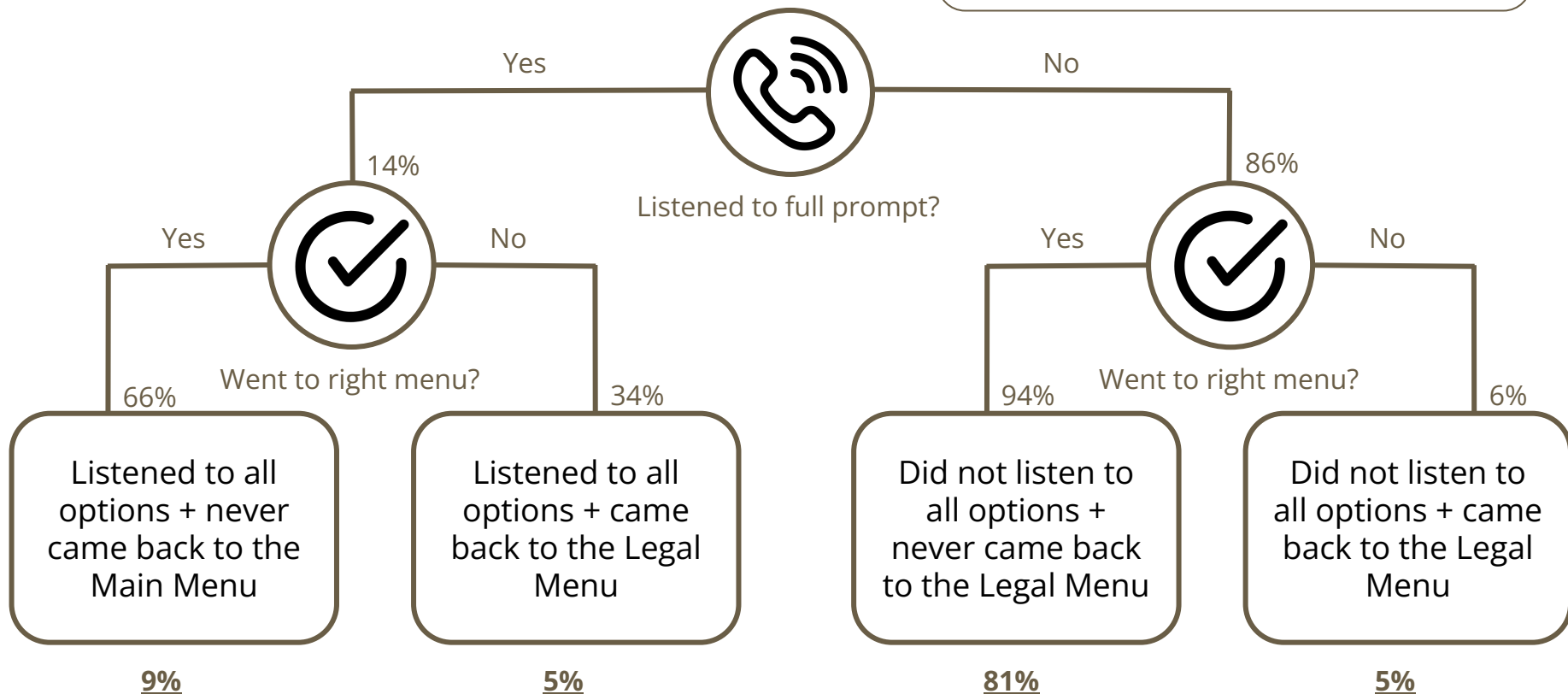
- The caller reached or entered a queue

Algorithm

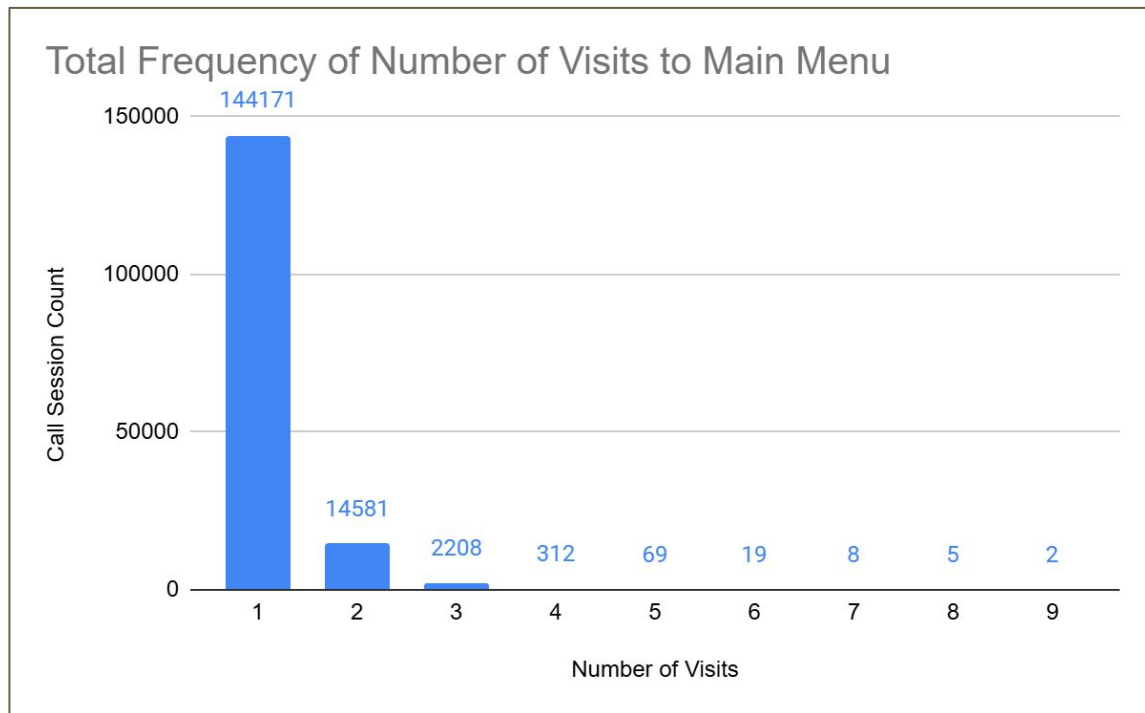
- Check if session reaches "ClosedQueueMenu" or a non empty "Queue Name"

Main Menu Analysis

Approximately **81%** of callers do not listen to the full prompt and never return to hear the prompt again



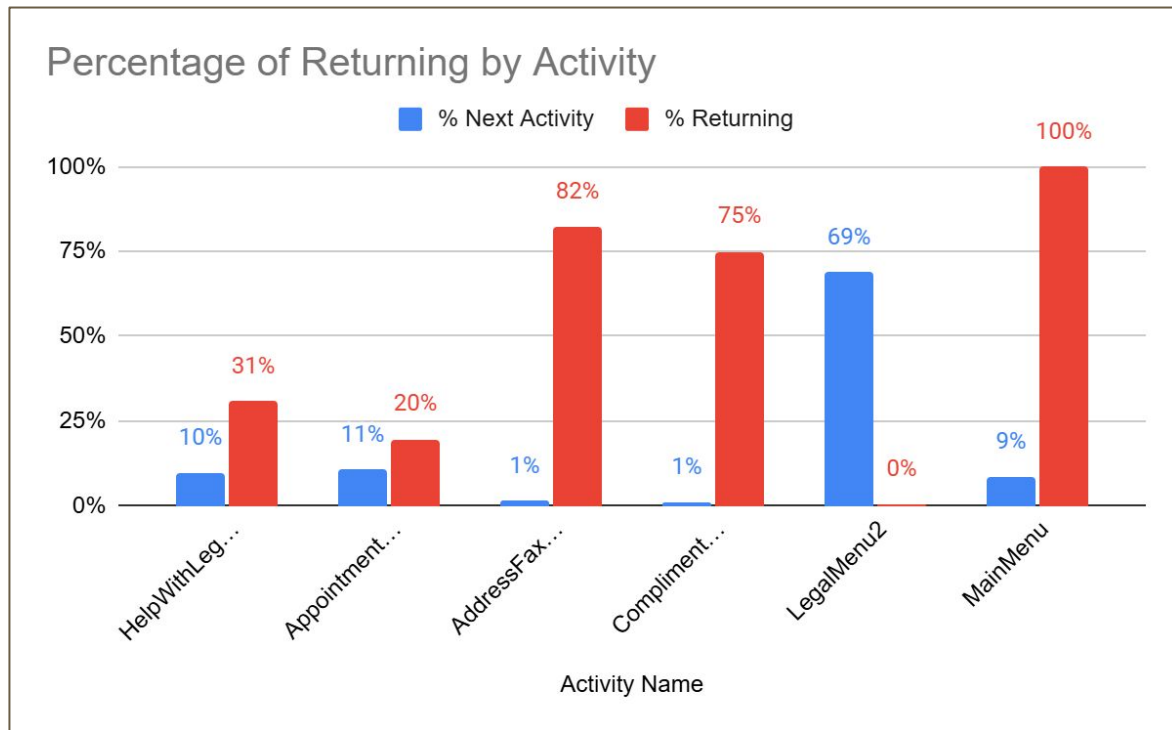
Main Menu Analysis Continued



Approximately **50%** of cases of 2+ Main Menu counts were consecutive repeats → prompt repeated at the node

Approximately **50%** of calls that reached the Main menu reached a queue → successful call

Main Menu sub-menu Analysis



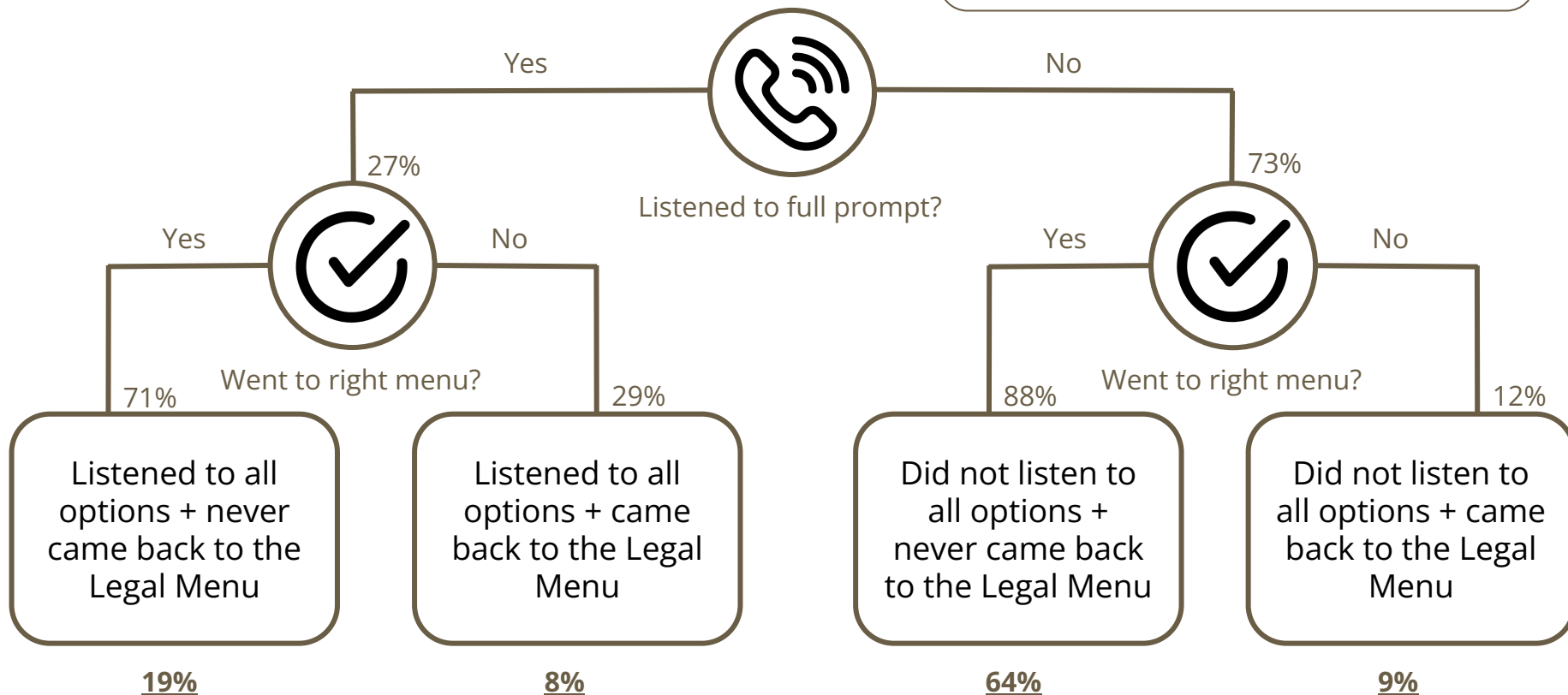
6 possible activities that have an existing path back to the MainMenu

Majority of calls proceeded to LegalMenu2 (69%), yet the return rate is near 0%

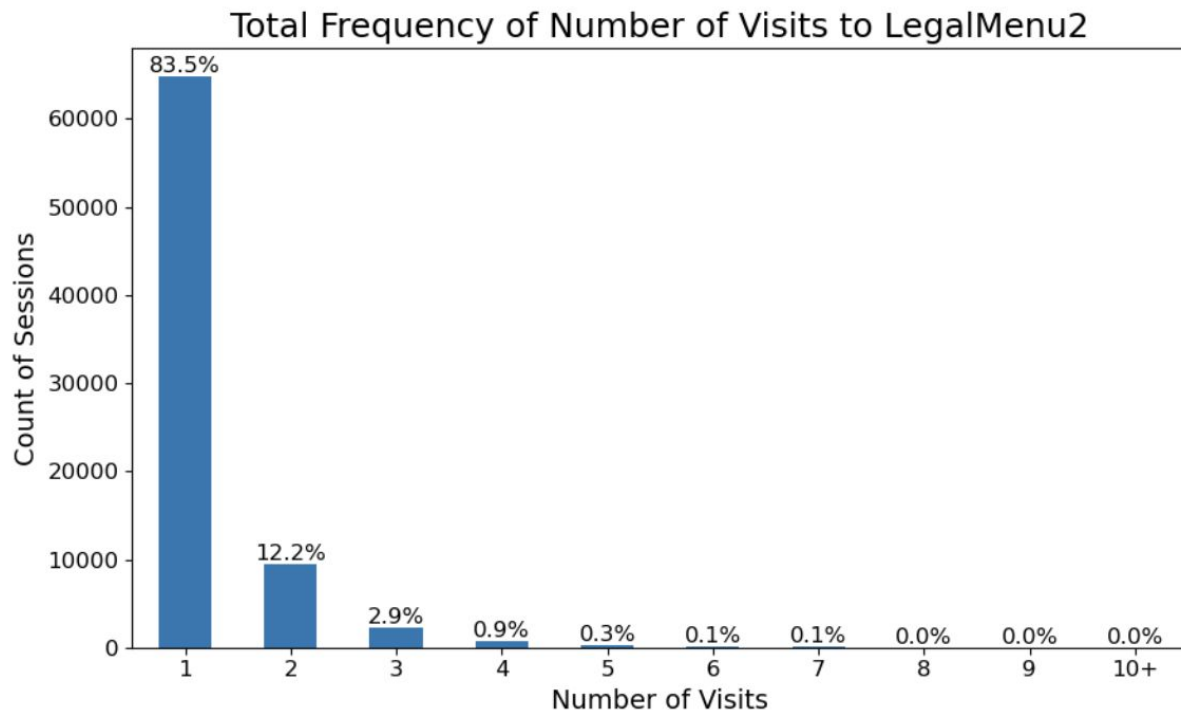
→ Little misguidance from MainMenu to LegalMenu2

Legal Menu Analysis

Approximately **64%** of callers do not listen to the full prompt and never return to hear the prompt again



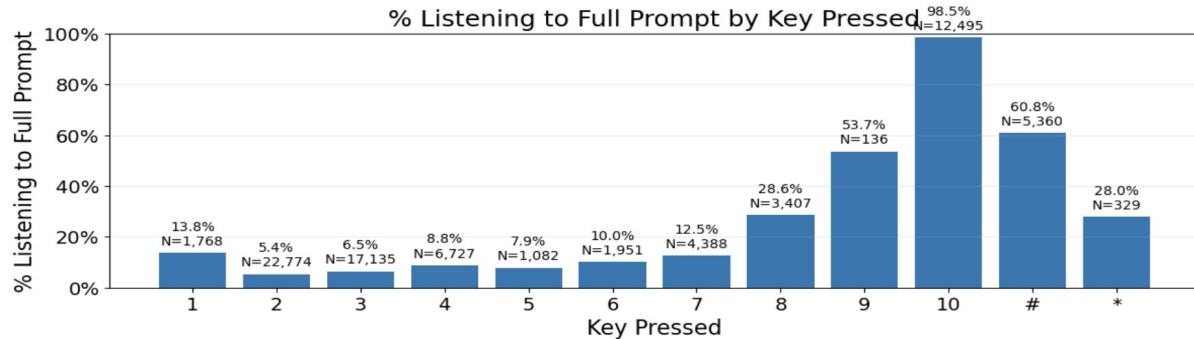
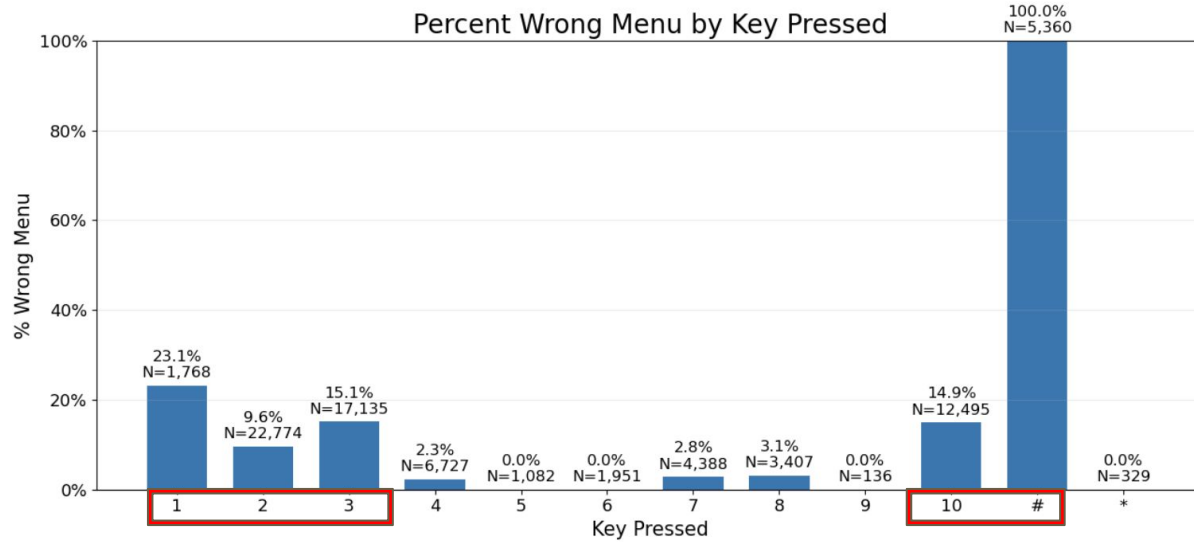
Legal Menu Analysis Continued



Approximately **40%** of cases of 2+ Legal Menu counts were consecutive repeats → prompt repeated at the node

Approximately **65%** of calls that reached the legal menu reached a queue → successful call

Legal Menu sub-menu Analysis

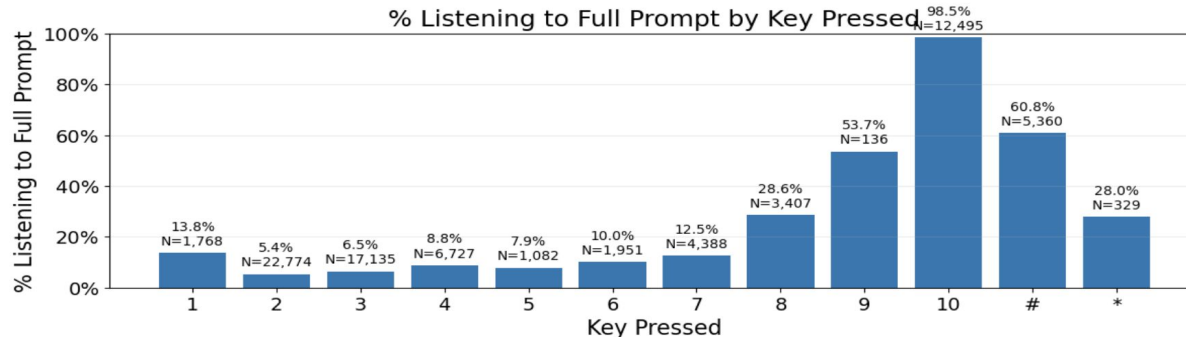
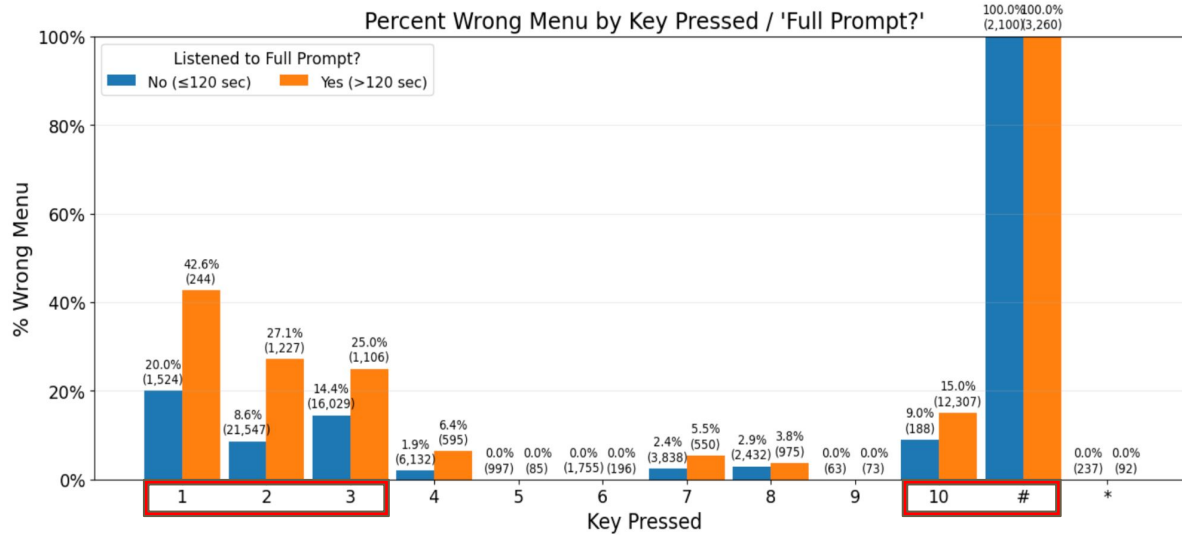


1. HIV
2. Family
3. Housing
4. Benefits
5. Consumer(SP) Queue
6. Criminal Records Voicemail
7. Employment
8. Immigration
9. ADAPT(SP) Queue
10. Other Legal

#/no/invalid input.
Legal Menu (again)

*. Front Desk Transfer

Legal Menu sub-menu Analysis



1. HIV
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Legal Menu Further Steps

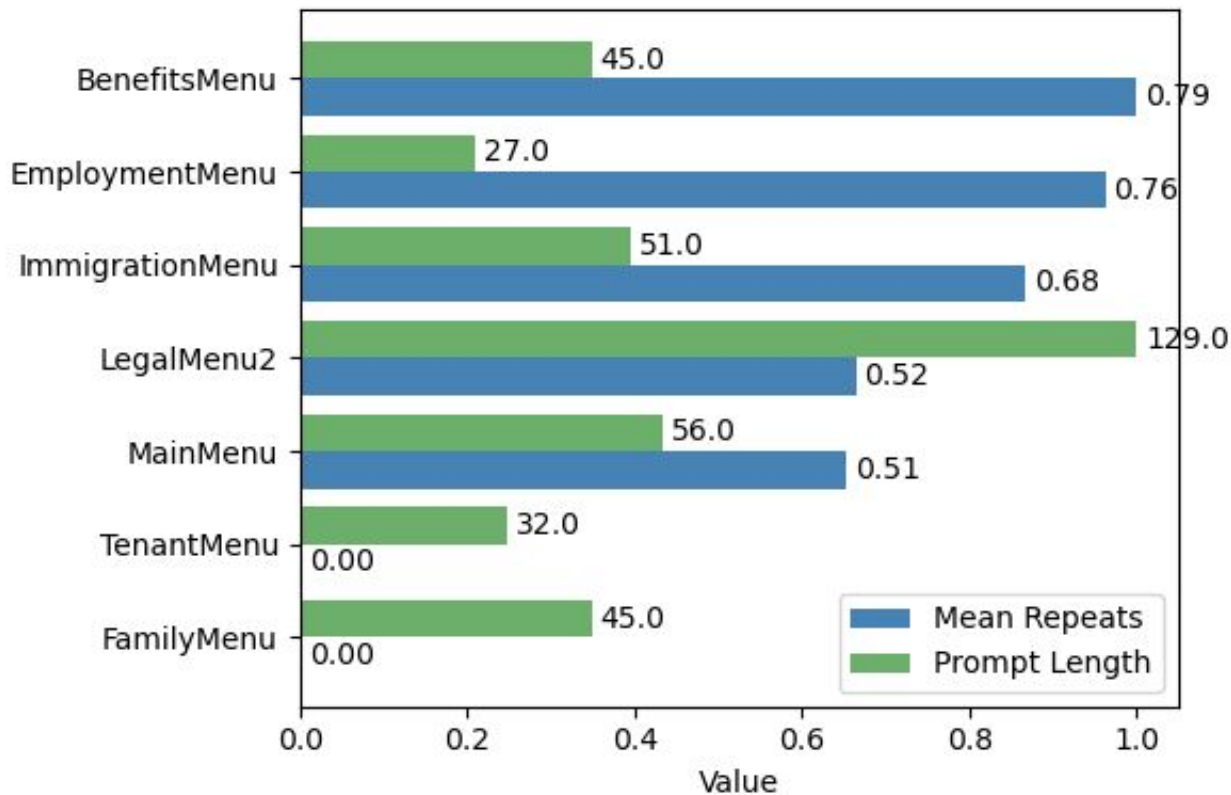
- Early options (HIV, Family, Housing) have disproportionate number of revisits to legal menu, despite most callers not listening to full set of options (likely verbiage issue and issue with ordering / confusion with a later submenu option)
 - Goal: Analyze the submenus chosen on second / third visit to legal menu
 - Do callers confuse two sub-menus?
 - Do callers choose an option too early?
 - Which submenus do callers choose after directly repeating the legal menu prompt?
- In addition to '% right submenu' and '% listening to full prompt' as variables, introducing third variable ('% eventually reaching queue', i.e. successful call journey)
 - Goal: Analyze impact of multiple visits to legal menu on eventual call success
 - High percentage reaching queue after second legal menu prompt (especially for early options), may indicate the potential effectiveness of verbiage improvement
 - I.e. whether the callers initially reaching wrong sub-menu can be matched to their correct sub-menu (or the caller's issue does not apply to any sub-menu / legal aid cannot help)

Prompt Repetition Analysis

- Based on our previous analysis, a significant portion of repetitions occur at the node level (repeating the prompt, rather than circling back to the prompt)
- To identify these, I analyze two measures (mean repeats and redundancy ratio) across multiple prompts.
- We focus only on callers who have listened to the full prompt, remain in the same menu node, and do not navigate to another menu before potentially repeating the prompt → isolate to node level delay.
- Analysis is limited to prompts with >3 options that allow repeating the prompt.

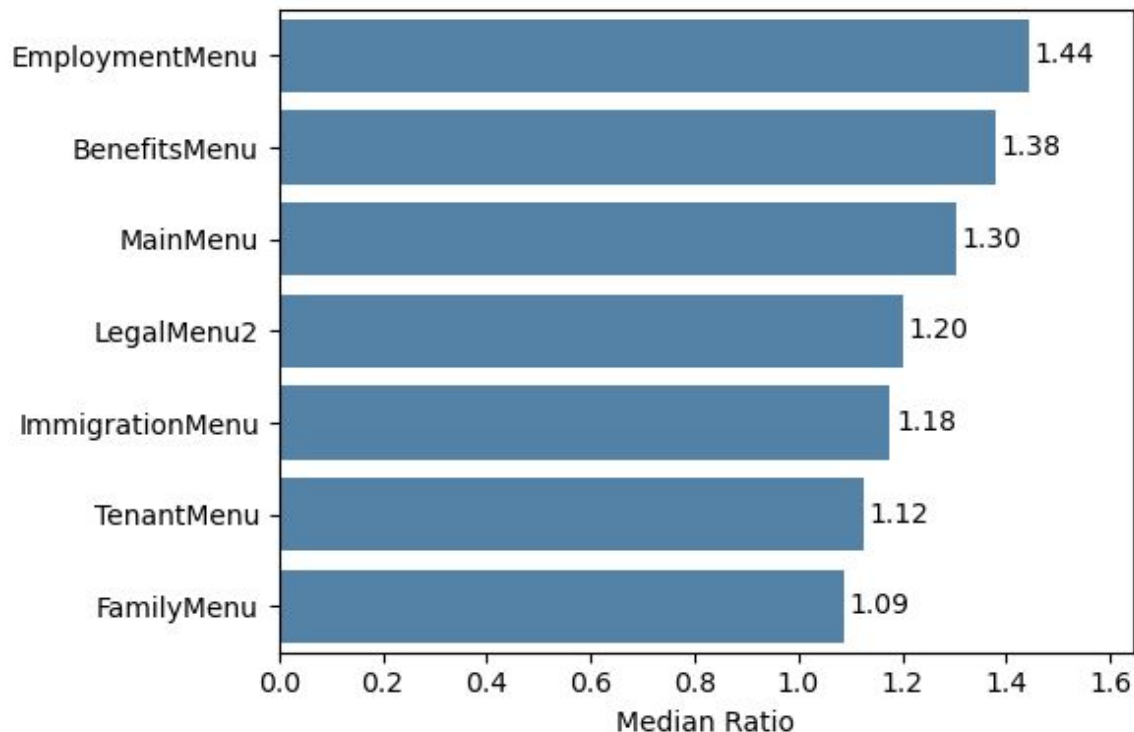
Mean number of repeats vs prompt length by Prompt

*outliers removed



- Repeats come from hitting the pound key, invalid inputs, or no input, all strong indicators of confusion.

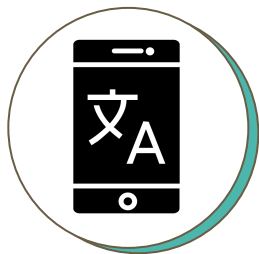
Median redundancy ratio by Prompt (n > 900 for all categories)



- This shows prompts that are causing disproportionate delay **relative to their length**.
- Currently noisy due to inability to distinguish between spanish/english prompts

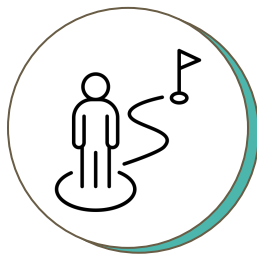
- **Redundancy ratio = total time / prompt length**

Next Steps/Remaining Challenges



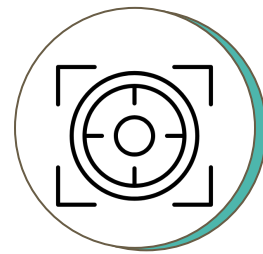
CAR data does not differentiate language

Identifying English vs. Spanish Prompts necessary for more precise time analysis



Incorporating “successful call” variable into our analysis

Useful to determine the impact of multiple visits on entering a queue



More granular approach in analyzing when people are leaving in a prompt to analyze decisions of second, third, ... visit

Thank You!
